United International **University** (**UIU**)
Dept. of Computer Science and Engineering (CSE)
Mid Term Assessment Year: **2025** Semester: **Summer**
Course: **CSE 323/CSE 3711/EEE 4413** Title: **Computer Networks**
Marks: **30** Time: **1 Hour 30 minutes**

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

There are **3 (Three)** questions. Answer **all 3 (Three)** questions. All questions are of values indicated on the right-hand margin.

Q1. a) What are the **four** key topologies used in networking? Explain with necessary diagrams and label those correctly. **[2 + 4 = 6]**

b) For TCP/IP model, what is the port address? Mention an example of port address. **[1 + 1 = 2]**

c) Which layer of OSI reference model is responsible for *dialog control* and *synchronization*? Draw an operational diagram of that layer. **[1 + 2 = 3]**

d) What is a protocol for communication between two nodes? Describe the three key elements of a protocol. **[3]**

Q2. a) If we consider the network structure of the Internet, what are the **three** key parts of it? (Hint: One part is "*Network Edge*"). Draw a diagram and explain. **[4]**

b) What is the difference between ***guided*** and ***unguided*** media? Why are they called so? Give examples of each. **[2]**

c) Suppose **10 cars** are traveling together as a caravan, following each other in a fixed order. Think of **each car** as a **bit** and **the caravan** as a **packet**. Also, suppose that each toll booth services (that is, transmits) a car (1 bit in this case) at a rate of one car per **30 seconds**, and that it is late at night so that the caravan's cars are the only cars on the highway.

Now, let us consider that a car can propagate at **100 km/hr**. The scenario is shown in **Fig. 1**. If the caravan is lined up before 2nd toll booth after **50 minutes**, what is the distance between the two toll booths? (Show all the necessary calculations and provide explanation. Simply stating the result will not be given any mark.) **[2]**

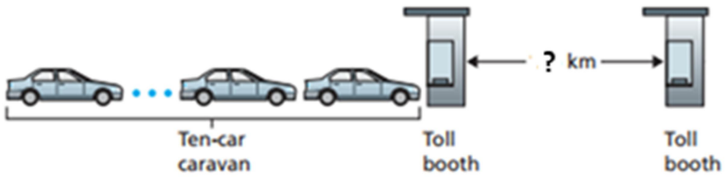


Fig. 1.

Q3. a) Define **Link Bandwidth**, **Bandwidth**, and **Throughput**. What units are used for these? **[3 + 1 = 4]**

b) What is a network app? How does it differ from other apps? **[2]**

c) Describe the key characteristics of **P2P architecture**. **[2]**

---------- **END of Question** ----------

Key characteristics of P2P architecture
No always-on server
Peers communicate directly
Each peer acts as both client and server
Self-scalable
Peers may join and leave at any time
Complex management